

Shuhan Zheng

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Summary

M.Eng. graduate from the University of Toronto. Demonstrated strong experience in streamlining complex analytical problem solutions via approaches grounded in first principles. I bring expertise in physics-aware software development, especially pertaining to nonlinear behaviours in dynamical systems, or ones requiring intensive computation. Multidisciplinary engineering, large data-driven projects, and complex cross-system work also fall within my forte. Supplementary skills in electronics, mechanical prototyping, and systems-level analysis.

Education

University of Toronto Institute for Aerospace Studies (UTIAS) 2024–2026
M.Eng. in Aerospace Engineering
• Coursework: Spacecraft Dynamics and Control, Control Theory, Computational Fluid Dynamics, Project Management, Operations Management, etc.

University of Toronto 2020–2024
Honours B.Sc. with double majors in Physics and in Astronomy & Astrophysics

Experience

Research Assistant / Supervised Research Project Spring 2024–present
University of Toronto Institute for Aerospace Studies, Aerospace Mechatronics Lab
• Authored paper: *Dichotomizing Temporarily Bound Objects Using Capture States*. Abstract accepted by AMOS 2026 conference.
• Space Domain Awareness study powered by techniques such as statistical modelling and probabilistic neural networks, enabling rapid and early detection of mission-viable asteroids.

Research Assistant 2023–2024
University of Toronto, Near Field Cosmology Group
• Led the development of a **turnkey solution** for automatic retrieval, sanitising, and cataloguing of stellar data, with a portable interface for machine-learning model ingestion.
• Project was adopted by the Near Field Cosmology Group; the resulting database supports subsequent publications.

Research Assistant Intern Summer 2023
China Spallation Neutron Source, Chinese Academy of Sciences
• Led the development of a Python-based deterministic simulation tool for generating synthetic polarised-neutron images of magnetised samples.
• Reduced computational time from days to hours compared to conventional Monte Carlo methods.

Space Systems Member 2021–2023
University of Toronto Aerospace Team
• Co-authored team design papers for the 2022 and 2023 Small Satellite Conferences as a member of the FINCH mission team.
• Implemented and tested image-correction algorithms for hyperspectral remote-sensing data in collaboration with the optics team. The algorithms will support the upcoming FINCH **microsatellite mission**.

Vice-President, Internal and External Affairs 2023–2024
University of Toronto Physics Student Union

- Managed the organisation, logistics, and coordination of outreach, fundraising, social, and academic events.
- Acted as a liaison between faculty and students.

OPS Safety Operator

2026

Launch Canada

- Responsible for assisting the launch OPS and Safety team, enforcing launch pad clearance and compliance with launch safety regulations.

Projects

AMAT – Agentic Mission Analysis Toolbox

Independent Software Project (Link: [GitHub](#))

- Independently built a modular software platform with swappable backend physics engines and trajectory libraries, for deterministic spacecraft mission analysis and flight planning. Supports GMAT, SPICE, Orekit, and more.
- Interface for onboard or ground-based intelligence.

Stability and Station-Keeping Near the Earth-Moon L_2 in Planar CRTBP

Nonlinear Control Theory Course Paper

- Modelled spacecraft motion near the Earth-Moon L_2 point using the Circular Restricted Three-Body Problem, then assessed the stability and station-keeping requirements.
- Implemented Python simulations and designed an LQR controller that stabilized perturbed trajectories, demonstrating closed-loop convergence near an otherwise unstable orbital equilibrium.

GMAT Automation for Asteroid Trajectory Prediction

Orbital Dynamics Course Project (Link: [GitHub](#))

- Individual study on Monte Carlo trajectory prediction of asteroid-Earth encounters.
- Predictions were verified against published journal literature.

Catalogue Acquisition and Compilation Tool

Astronomy Capstone Project (Link: [GitHub](#))

- Independently built an autonomous data retrieval, sanitising, and management pipeline with user-oriented API.
- Scalable scientific database solution designed for hierarchical and relational data.

Polarisation-modulated Optical Communication Demonstrator

Electronics Course Project

- Designed and built an optical-electronic prototype for communication via a laser link.

Publications

- Miles, A., Maranto, D., Xu, S., ..., **Zheng, S.**, et al. (2022). *FINCH: A Blueprint for Accessible and Scientifically Valuable Remote Sensing Satellite Missions*. Small Satellite Conference, Weekend Session 7: Science/Mission Payloads – Research & Academia II, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2022/all2022/88/>
- Vyse, I., Dagli, R., Chadha, D. V., Ma, J. P., Chen, H., ..., **Zheng, S.**, & Nguyen, K. (2024). *Beyond the Visible: Jointly Attending to Spectral and Spatial Dimensions with HSI-Diffusion for the FINCH Spacecraft*. Small Satellite Conference, Session X: Ground Systems, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2024/all2024/121/>

Skills

Programming and Analysis: MATLAB, LaTeX

Aerospace and Simulation Tools: GMAT, SPICE, COMSOL Multiphysics, SolidWorks, CFD